Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

First Semester B.Tech Degree Regular Examination December 2024 (2024 Scheme)

Course Code: GAPHT121

Course Name: PHYSICS FOR INFORMATION SCIENCE

Max. Marks: 60 Duration: 2 hours 30 minutes

PART A

(Answer all questions. Each question carries 3 marks)

		CO	Marks
1	Write any three important postulates of classical free electron theory.	CO1	(3)
2	What is meant by critical field in superconductors? Explain the variation of critical field with temperature.	CO1	(3)
3	Explain the natural broadening of spectral lines emitted from an exited atom by using Heisenberg's uncertainty principle.	CO2	(3)
4	Explain the phenomena of Quantum mechanical tunnelling.	CO2	(3)
5	What are extrinsic semiconductors? How are they formed?	CO3	(3)
6	What is biasing? Explain the effects of forward biasing in a diode.	CO3	(3)
7	Explain the process of Zener breakdown in diodes.	CO4	(3)
8	How light is produced in an LED? Give any two advantages of LEDs over other conventional lamps.	CO4	(3)

PART B

(Answer any one full question from each module, each question carries 9 marks)

Module -1

9	a)	Obtain the expressions for current density and electrical conductivity of a metallic conductor in terms of the concentration of free electrons in it.	CO1	(6)
	b)	In a metal, consider the energy level lying 0.01eV above the Fermi level. What is the probability of this level being occupied by an electron at 300K. Given Boltzmann constant= 1.38×10^{-23} J/K.	CO1	(3)
10	a)	Compare Type I and Type II superconductors with proper figures. Write two examples for both.	CO1	(6)
	b)	Write six applications of superconductors in various fields.	CO1	(3)

Module -2

11	a)	What is a quantum mechanical wavefunction? Explain its important characteristics.	CO2	(3)
	b)	Starting from the equation of a plane wave, derive time dependent Schrodinger's wave equation.	CO2	(6)
12	a)	Derive the expression for the eigen function for a particle in one dimensional infinite square well potential.	CO2	(6)
	b)	Find the two lowest energy levels of an electron in a one dimensional infinite square well potential of width $3A^0$.	CO2	(3)

Module -3

13	a)	Derive an expression for the density of electrons in the conduction band of an intrinsic semiconductor.	CO3	(6)
	b)	Prove that the location of fermi level in an intrinsic semiconductor at 0K lies at the middle of valance and conduction bands.	CO3	(3)
14	a)	Explain the formation of a p-n junction by drawing the energy band diagram of a p-n junction at equilibrium condition. Mark the fermi level and the directions of diffusion and drift motions of charge carriers across the junction.	CO3	(6)
	b)	Calculate the diode current in a silicon diode at 25^0C when a forward bias voltage of 0.5V is applied. The reverse saturation current at 25^0C is 0.1μA (material constant, η =2).	CO3	(3)

Module -4

15		Explain the centre-tap and bridge type full wave rectifiers with proper diagrams and obtain its efficiency and ripple factor.	CO4	(9)
16	a)	What is a photo diode? Explain the construction and working of a PIN photodiode.	CO4	(5)
	b)	Write any four applications of photodiodes.	CO4	(4)

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